



Self-Improvement

$$\mathcal{A}_{t+1} = \text{IMPROVE}(\mathcal{A}_t; \mathcal{S}_t), \quad \mathcal{A}_t = (\theta_t, \Sigma_t), \quad \Sigma_t := (p_t, m_t, \mathcal{T}_t, g_t)$$

(1) FM Improvement

$$\theta_{t+1} = \text{IMPROVE}_{\theta}(\theta_t; \mathcal{S}_t), \quad \Sigma_{t+1} = \Sigma_t$$

Key Improvement: $\theta_t \rightarrow \theta_{t+1}$

Self-generated Data

- Generation strategies
- Data Formats
- Application Domains

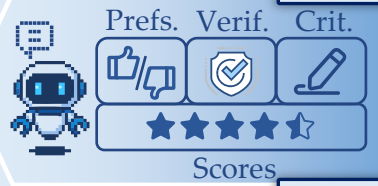
$$\mathcal{S}_t \approx \mathcal{D}_t$$



Self-generated Supervision

- Feedback via Outcome Verification
- Feedback via Self-Consistency
- Feedback via Self-Refinement

$$\mathcal{S}_t \approx r_t$$



Self-generated Experience

- Environment-Interaction
- World Models

$$\mathcal{S}_t \approx \mathcal{T}_t$$



SFT & RL

SFT & RL

(2) Scaffolding Improvement

$$\Sigma_{t+1} = \text{IMPROVE}_{\Sigma}(\Sigma_t; \mathcal{S}_t), \quad \theta_{t+1} = \theta_t$$

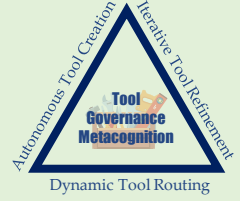
Key Improvement: $\Sigma_t \rightarrow \Sigma_{t+1}$

Prompt Optimization

- Scalar-Feedback Optimization
- Qualitative-Feedback Refinement
- Population-Based Evolution
- Gradient-Based Optimization

$$p_{t+1} = \text{IMPROVE}_p(p_t; \mathcal{S}_t)$$

Tool



$$\mathcal{T}_{t+1} = \text{IMPROVE}_{\mathcal{T}}(\mathcal{T}_t; \mathcal{S}_t)$$

Memory

- Memory Object
- Memory Structure
- Memory Processing

C R U D

$$m_{t+1} = \text{IMPROVE}_m(m_t; \mathcal{S}_t)$$

Full Scaffolding

- Self-Referential Code Update
- Generate-Test-Patch Loop
- Open-Ended Search over Agent Designs

$$\Sigma_{t+1} = \text{IMPROVE}_{\Sigma}(\Sigma_t; \mathcal{S}_t)$$

Self-Improvement AI Agent

A Survey



Software Engineering



Web Navigation and Automation



Gaming and Strategic Reason



Scientific Discovery



Embodied AI and Robotics



General Computer Control